

Experiment 2: Separation of mixtures by filtration and crystallization

The separation of a mixture of two solids can often be achieved by filtration and crystallisation. To be successful, this requires that the components of the mixture have different solubilities in a particular solvent. The purpose of this experiment is to separate sodium chloride/charcoal and sodium chloride/copper(II) chloride mixtures.

Equipment

filter funnel stand	filter funnel
Bunsen, tripod and gauze mat	glass rod
beakers (two 100 mL)	graduated cylinder (25 mL)
boiling chip	ethanol (25 mL)
distilled water	watch glass
filter paper (Whatman No. 1-three 12.5 cm sheets)	
sodium chloride/charcoal mixture (4 g)	
sodium chloride/copper(II) chloride mixture (8 g)	

Procedure

A. Separation of a Sodium Chloride and Charcoal Mixture

1. Place 4 g of the salt/charcoal mixture in a 100 mL beaker and add about 15 mL of distilled water. Stir the mixture for about two minutes to allow the salt to dissolve.
2. Set up a filter funnel with filter paper on a filter funnel stand. Filter the mixture and collect the filtrate in a 100 mL beaker as shown in Figure 1.1.
3. Wash the solid with a further 5 mL of water but do not add this to the filtrate. Note and record the appearance of the solid.
4. Add a boiling chip to the filtrate. Heat the solution with a bunsen burner and boil gently to reduce the volume.
5. When crystals of sodium chloride appear, turn off the bunsen and allow the solution to cool.
6. Record the appearance of the sodium chloride crystals.

B Partial Separation of a Sodium Chloride and Copper(II) Chloride Mixture

1. Place 4 g of the sodium chloride/copper(II) chloride mixture in a 100 mL beaker and dissolve in about 15 mL of distilled water.
2. Warm the solution with a bunsen and boil gently until crystals begin to appear in the solution. Remove the bunsen and cool the solution.
3. Filter the solid formed and wash with about 5 mL of ethanol. Record the appearance of the solid and the filtrate.

C Separation of a Sodium Chloride and Copper(II) Chloride Mixture

1. Weigh out approximately 2 g of sodium chloride and record its mass accurately. Add this to a 100 mL beaker.
2. Weigh out approximately 2 g of copper(II) chloride and record its mass accurately. Add this to the same 100 mL beaker.
3. Add about 15 mL of ethanol to the beaker and stir the mixture for about two minutes.
4. Weigh a piece of filter paper.

Notes

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5. Filter off the undissolved solid using the weighed filter paper. Wash this solid with about 5 mL of ethanol. Note the appearance of the solid.
6. Allow the solid in the filter paper to dry completely and reweigh. Calculate the mass of solid in the filter paper.
7. Place about 5 mL of the filtrate on a watch glass and allow the ethanol to evaporate. Record the appearance of the crystals that form.

Processing of results and questions

1. What property enabled you to completely separate the sodium chloride/charcoal mixture?
2. In part B you were able to separate some pure sodium chloride but the filtrate contained both sodium chloride and copper(II) chloride. What can you say about the solubility's of these two compounds in water?
3. In part C you were able to more completely separate the sodium chloride and the copper (II) chloride using ethanol as a solvent.
 - a. Explain why this was possible using ethanol but not when water was used as a solvent in part B.
 - b. Calculate the percentage of sodium chloride recovered from the mass of solid in the weighed filter paper
 - c. Describe a systematic error in this experimental design that would result in less than 100% recovery of the sodium chloride from the initial mixture.